

The Internet today has penetrated multiple devices, which are now present in multiple form factors, be it smart watches, mobile phones, tablets or even TVs. The varied form factors have definitely changed the way we interact with our computing devices and have taken us in the direction of natural interfaces. Organisations are now trying to evolve newer ways of making interaction with computers much more easier and much more natural. Gone are the days when a keyboard, mouse and dot matrix printers were the only ways to get data in and out of the system. Intel is one such company, which has always been at the crux of the computer manufacturing industry and has helped drive computer manufacturers into developing new ways of interacting with computers.

Intel's RealSense technology extrapolates the existing devices to involve not just you, but also all your surroundings and you realise just how much computers have evolved to now simultaneously see, hear and feel, introducing new ways to process this information and revolutionise the way we use them. This is expected to not only make the computing experience more immersive, but also shrink the line between the real and virtual world for you. Some features such as speech and facial recognition, finger sensing, gesture recognition and 3D scanning/printing bring out the third dimension from our computers. This allows you to naturally manipulate and play with 3D objects and interact with them as if you were playing or working with your buddy sitting next to you.

While many may feel that these technologies are still a few years away from reaching the market, they're here and developers are already bringing out newer applications of these features as we speak. The only thing required now is to have developers build and innovate from what we already have and take it forward to newer heights. It's only when this technology comes into the hands of mainstream users will it spur further innovation to make the experience even better. At present, it needs to be proven that there's a real need for these devices and that consumers should upgrade to newer devices. Some companies have already made this interactive technology

mainstream in the form of devices such as Kinect and Wii U, but its application has mostly been restricted to casual gaming. As the market growth for PCs continues to decline, innovation in this direction will definitely generate a renewed vigor in the purchase of personal computing devices.

Intel has been working to introduce something very similar to the mainstream computing devices, albeit in a different and exceptional way. Its journey in this domain began with the Perceptual Computing SDK, which was released in 2013. The RealSense SDK is somewhat of a sequel to the same chapter with some tweaks and many new features. The Beta version of this SDK is due to be released in Q3 of this year.

REALSENSE SDK: THE FEATURES

The Intel® RealSense™ SDK is a software development kit available for systems based on Intel Core processors 4th Generation and onwards. It will be available for download starting Q3 in 2014, but developers can sign up now on the Intel website (<http://goo.gl/>

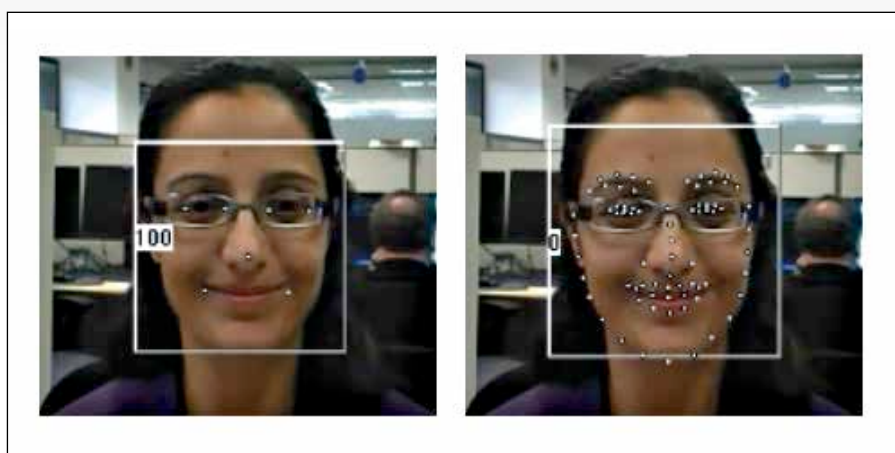
before. Some of the key technology areas in which this SDK works are as follows:

SPEECH RECOGNITION

In partnership with Nuance, Intel has released Speech Recognition features for the RealSense SDK. These features work with the help of Nuance Dragon Assistant, a well known software stack available for natural language processing applications on Windows 8. It works on top of the Microsoft Speech API and enables the user to speak into the microphone and directly interact with the computer as if he was speaking to the application in a conversational manner. This is made possible by the contextual dialogue capabilities of the system. It provides integration with more than 170 content providers including Spotify and TripAdvisor, among others. This allows you to get voice-activated directions, flight bookings or dinner reservations directly from within your apps.

FACIAL ANALYSIS

Facial Analysis features have taken a turn for the better in this release of the RealSense SDK. The features now support depth with 78



The Facial Analysis module supports 78 landmark points for better accuracy

hMOLF) to be notified when it's available for download. The SDK allows easier access to all the features supported by the RealSense technology. With the help of this software stack, developers can now deploy next generation applications with natural interfaces while concurrently bringing out the best in the collaborative features in a way never possible

landmark points for accuracy like never before and enable 3D detection of the face with roll, pitch and yaw. This is a big improvement in comparison to the 7 landmark points supported in 2013. It can also identify specific facial features and the direction in which your face is turned to provide useful contextual feedback to your application. What's more, it can even